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Research Article

CBE Clima Tool: A free and open-source web application for climate analysis tailored to sustainable building design

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Abstract

Climate-responsive building design holds immense potential for enhancing comfort, energy efficiency, and environmental sustainability. However, many social, cultural, and economic obstacles might prevent the wide adoption of designing climate-adapted buildings. One of these obstacles can be removed by enabling practitioners to easily access, visualize and analyze local climate data. The CBE Clima Tool (Clima) is a free and open-source web application that offers easy access to publicly available weather files and has been created for building energy simulation and design. It provides a series of interactive visualizations of the variables contained in the EnergyPlus Weather Files and several derived ones like the UTCI or the adaptive comfort indices. It is aimed at students, educators, and practitioners in the architecture and engineering fields. Since its inception, Clima's user base has exhibited robust growth, attracting over 25,000 unique users annually from across 70 countries. Our tool is poised to revolutionize climate-adaptive building design, transcending geographical boundaries and fostering innovation in the architecture and engineering fields.

Keywords

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climate analysis
sustainable building design
web application
building energy simulation
open-source software

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1 Introduction

Buildings construction and operation account for approximately 36% of global final energy use and nearly 40% of energy-related carbon dioxide (CO_2) emissions (United Nations 2018). However, according to a survey conducted on over 90,000 individuals in approximately 900 office buildings in North America, 39% of the participants were not satisfied with their thermal environment, 34% with noise, 31% with air quality and 13% with lighting (Graham et al. 2021; Parkinson et al. 2023). There is a need to reduce greenhouse gas emissions related to buildings and increase their resilience while improving indoor environmental quality. Energy consumption can be reduced and occupant comfort can be improved when a building is climatically adapted to its environment (Brager and Baker 2009; Day and Gunderson 2015; Altomonte et al. 2019). The climatically adapted design of a building must take into account the outdoor and indoor climates, optimize site selection, building orientation relative to the sun and the wind, daylight, and view access (Ratti et al. 2003; Santamouris and Kolokotsa 2013; Ko et al. 2022). This process is present in vernacular design traditions all over the world (Mileto et al. 2014).

1.1 Early efforts in climate-adaptive design

In the 19th century, the early modern movement in architecture took a renewed interest in questions related to solar exposure and orientation, as can be recognized in many sketches, built works, and theoretical writings by Le Corbusier and Alvar Aalto (Menin and Samuel 2003). After World War II, data and physics-based approaches to building climate adaptation emerged. In the early 1950s, the Form and Climate Research Group at the Columbia Graduate School of Architecture sought to develop techniques for refining the design process according to climatic adaptability (Barber 2020). Almost contemporary, the American Institute of Architects started publishing a seminal set of climate

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List of symbols

CSV	comma-separated values	SVG	scalable vector graphics
EPW	EnergyPlus Weather	T_{db}	dry-bulb air temperature (°C)
GHI	global horizontal irradiance (Wh/m^2)	U	wind speed (m/s)
RH	relative humidity (%)	UTCI	universal thermal climate index (°C)

graphs to describe the climate of different American cities (Taylor and Coe 1949).

1.2 Advancements in climate data visualization tools

The work about climate data, climatic conditions, and their implication for architectural design was then brought forward by the research of Olgyay (1963), among whose many contributions there is the Bioclimatic chart. Shortly after, the diffusion of Computer-Aided Design generated a few early tools that aimed at creating climatic data visualizations for architects and building engineers, such as the work by Murray Milne at UCLA (Milne 1978). From 1978 onward, Milne (1978) has developed several tools that use climatic data as design guidance, most notably Climate Consultant (Clayton et al. 1988). Figure 1 shows an example of the visualisations that can be generated using Climate Consultant. One of the early challenges for the Architecture Engineering and Construction (AEC) industry was the lack of a consistent climate data format. This issue was solved with the introduction of the EnergyPlus Weather (EPW) format (Crawley et al. 1999). In the early 2000s, Andrew Marsh developed Ecotect and the Weather Tool, now retired, which have enabled thousands of practitioners and students to integrate climate data in their CAD workflows for the first time (Roberts and Marsh 2001).

1.3 Current tools and challenges

The Weather Tool shared with Climate Consultant the fact of being a stand-alone, easyto-use application that enabled the visualization of weather data with some degree of customization. Recently, new tools have emerged in the field of climate data visualization for the AEC industry, most notably Ladybug Tools (Sadeghipour Roudsari and Pak 2013), Climate Studio (Solemma 2021), and cove.tool (cove.tool 2021). All of these projects enable different climate visualization directly in the most prevalent CAD environments, mainly Autodesk Revit and Rhinoceros 3D (McNeel et al. 2010). To achieve this integration, the tools act as software plug-ins. This strategy has certain advantages, i.e., showing contextual information in the user's design software of choice. On the other hand, it makes the tools part of a closed-source ecosystem and creates barriers to their use as they require multiple installations and increasing levels of specialization. For example, to use Ladybug one needs to be familiar with Rhinoceros 3d and its visual programming extension Grasshopper (McNeel et al. 2010). Existing stand-alone tools, like Climate Consultant, do not provide vector graphics and therefore their use in practice and education is limited (Steinfeld et al. 2012).

1.4 Contribution

We aim to develop open-source software that overcomes the limitations listed above and reduces barriers to access. The CBE Clima Tool represents a pioneering solution that redefines the landscape of climate analysis for sustainable building design. While various tools exist for calculating thermal comfort indices, its innovations in universal accessibility, user-friendliness, cross-platform compatibility, transparency, and community-based development position it as a valuable asset for professionals and researchers seeking to advance sustainable building design practices.

Universal Accessibility We are strongly committed to universal accessibility. Unlike traditional desktop applications, the CBE Clima Tool is web-based, providing architects, engineers, students, and educators worldwide with the ability to rapidly and reliably analyze and visualize complex climate data. This feature facilitates collaboration and knowledge exchange on a global scale.

User-Friendly Interface The user interface is characterized by its exceptional ease of use and interactivity. Users can seamlessly navigate the tool, effortlessly generating high-quality graphical representations of climate data. This user-friendly approach greatly enhances communication of climate-adapted building design concepts.

Cross-Platform Compatibility The web-based nature of our tool eliminates the need for local installations and ensures cross-platform compatibility. This accessibility ensures that users can use it regardless of their preferred operating system, further enhancing its practicality.

Transparency and Open Source Another key innovation is the CBE Clima Tool status as a free and open-source tool. This aspect fosters transparency in its code base, allowing users to review and validate the underlying algorithms and calculations. Such transparency builds trust in the tool's capabilities and encourages its adoption within the academic and engineering communities.

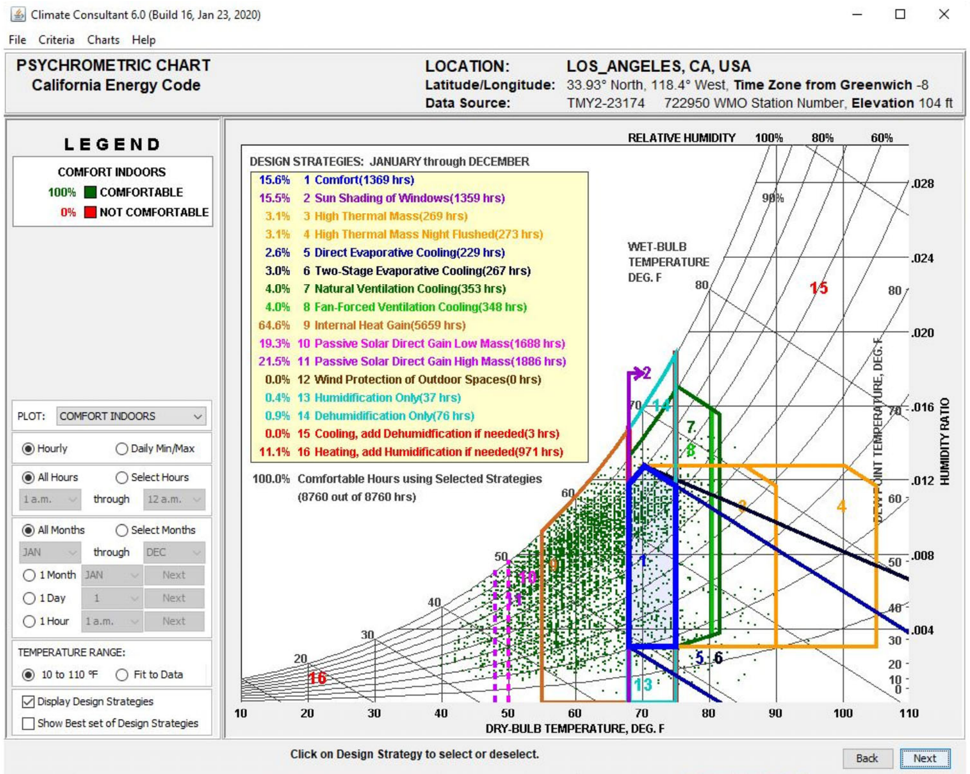

Fig. 1 Visualization of a weather file for the city of Los Angeles, CA, USA in the psychrometric chart. Sustainable strategies potential ranges of applicability are overlaid on the graph. This visualization was created with Climate Consultant 6.0

Community-Based Development The CBE Clima Tool not only benefits from open-source principles but also thrives on community-based development. This innovation encourages collaborative enhancements and ensures that the tool evolves with input from a diverse range of contributors. Moreover, it empowers users to incorporate our code into their applications, extending its impact and promoting knowledge sharing.

2 Software description

The CBE Clima Tool is a free, open-source web application for analyzing and visualizing climate data specifically designed to support the needs of architects, engineers, students, and educators interested in climate-adapted building design. It does not only visualize data, but it organizes, manipulates, and relates them in an easy-to-interpret fashion. The landing page is characterized by an interactive world map. The user can access circa 30,000 publicly available weather data files through this map. Data are collected from publicly available repositories maintained by EnergyPlus and Climate. OneBuilding.org. Alternatively, the user can upload any valid EPW file. The uploaded file is automatically processed and analyzed. Our tool does not perform quality checks other than checking that the data supplied matches the expected structure of an EPW file. An error message is shown to the user in case of a mismatch. The data contained within the EPW files are used to generate easy-to-interpret charts. Many of these allow users to explore data at different temporal resolutions, highlighting seasonal, daily, or hourly patterns. All the charts allow some degree of interactivity, allowing the user to hover, zoom, and visualize values and relevant statistics. For some charts, we allow the user to select specific time ranges (i.e., wind roses analysis). Not all relevant data for a climate analysis is contained directly in the EPW files but can be derived from the information contained therein. We use the open-source Python library pvlib (Holmgren et al. 2018) to calculate the apparent solar position (altitude and azimuth). We also use the open-source Python library pythermalcomfort (Tartarini and Schiavon 2020) to calculate the Universal Thermal Climate Index (UTCI) (Bröde et al. 2012) scenarios to understand outdoor comfort, solar gain on occupants (Arens et al. 2015), the adaptive thermal comfort model (ASHRAE 2020), natural ventilation potential, heating and cooling degree days, and various other psychrometric variables. The CBE Clima Tool can be accessed at this URL https://clima.cbe.berkeley.edu.

2.1 Software architecture

The CBE Clima Tool tool is built using Dash (Plotly Technologies Inc 2015), a Python framework to build web analytic applications and data visualization websites. We use Dash since it is an open-source library released under an MIT license, and applications built with it can be rendered in any web browser. Clima is, therefore, cross-platform and compatible with the most widely used web browsers. Dash applications are web servers that run Flask and communicate JSON packets over HTTP requests. We have deployed CBE Clima Tool using a cloud service that hosts our containerized application. We use Docker to create a container that listens to user requests. A request is triggered each time the user navigates to the URL https://clima.cbe.berkeley.edu. Dash's front-end renders components using React.js, a Javascript framework. While the back-end code runs on a cloud instance, the front-end code is run locally inside the user's browser. Each time a user changes some of the input parameters, a new request is sent to the server. The Python source code is stored inside the container, while the EPW data is queried from the databases maintained by EnergyPlus and Climate.OneBuilding.org. Figure 2 shows a diagram depicting the application architecture. For more information about how a Dash application works, please consult their official documentation at this URL https://cbe-berkeley.gitbook.io/clima.

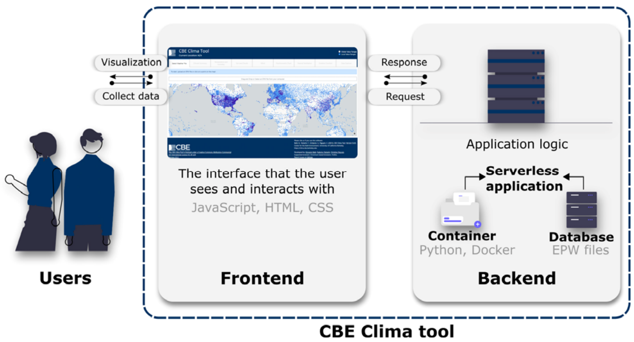

Fig. 2 Application architecture diagram describing the relationships between users and frontend and backend applications

2.2 Main features

The CBE Clima Tool allows users to analyze and visualize climatic data. Users can access EPW files via a map-based interface (as previously stated in Section 2). All images and charts included in our tool can be downloaded in Scalable Vector Graphics (SVG) format. This allows them to be modified and adapted according to the users' needs. Furthermore, the vector nature of the SVG format enables both high-resolution on-screen display and high-quality printing. This is an important tool feature as it enables visually compelling data communication with multiple stakeholders. The interactive nature of the graphs means that the user can inspect the underlying numerical information through rich mouse-over events. Users can download, in Comma-Separated Values (CSV) format, either the EPW file or the CBE Clima Tool data frame created in the app back-end to generate the plots. The latter contains the variables in the EPW and additional ones derived from it. The complete list of the variables contained in the Clima data frame can be found at the URL https://cbe-berkeley.gitbook.io/clima/documentation/tabs-explained/tab-summary/clima-dataframe. Table 1 and Table 2 provide an overview respectively of variables and graph types in CBE Clima Tool. The variables in Table 1 are divided between those that are directly imported from an EPW file and those calculated by our tool. Table 2 outlines all the graph types available in CBE Clima Tool, the tab to which they belong, the temporal resolution of the displayed data, and the variables that can be used to generate them.

When displaying the variables in the interactive plots, users can either select to use the Global or Local Value Ranges, using the radio button at the top of the page, see Figure 3. The "Global" option uses preset limits for the chart axes, chosen to cover the vast majority of the climatic ranges to be found on Earth. This allows users to compare charts generated for different locations effectively. The "Local" option sets the upper and lower limits of the chart

Table 1 List of variables available for climate analysis in CBE Clima Tool

EPW variables	Derived variables
Longitude	Humidity ratio
Latitude	Wet-bulb temperature
Hour	Solar Elevation
Day	Solar Azimuth
Month	Saturation pressure
Dry bulb temperature	Heating & Cooling Degree Days
Dew point temperature	Natural Ventilation Potential
Relative humidity	UTCI temperature (sun, wind)
Atmospheric pressure	UTCI temperature (no sun, wind)
Extraterrestrial horizontal radiation	UTCI temperature (sun, no wind)
Extraterrestrial direct normal radiation	UTCI temperature (no sun, no wind)
Horizontal infrared radiation	UTCI stress index (sun, wind)
Global horizontal radiation	UTCI stress index (no sun, wind)
Direct normal radiation	UTCI stress index (sun, no wind)
Diffuse horizontal radiation	UTCI stress index (no sun, no wind)
Global horizontal illuminance	
Direct normal illuminance	
Diffuse horizontal illuminance	
Zenith luminance	
Wind direction	
Wind speed	
Total sky cover	
Opaque sky cover	
Visibility	

Table 2 Graph types available in CBE Clima, the tab to which they belong, the temporal resolution of the displayed data and the variables that can be used to generate them

Graph type	Tab	Temporal resolution	Variables	Optional variables
Yearly profile	Temperature and humidity, data explorer	Daily	Any	—
Monthly profile	Temperature and humidity, sun and clouds, data explorer	Hourly, monthly	Any	—
Heatmap	Temperature and humidity, data explorer	Hourly	Any	—
Stacked bar	Climate summary, climate summary, sun and clouds, natural ventilation, outdoor comfort	Monthly	Heating degree days, cooling degree days, cloud cover, natural ventilation potential, UTCI heat stress index	—
Violin	Climate summary	Yearly	Dry bulb temperature, relative humidity, global horizontal radiation, wind speed	—
Spherical sun path	Sun and clouds	Hourly	Solar altitude, azimuth	Any
Cartesian sun path	Sun and clouds	Hourly	Solar altitude, azimuth	Any
Wind rose	Wind	Yearly, seasonal, custom	Wind speed, wind direction	—
Psychrometric chart	Psychrometric chart	Hourly	Dry bulb temperature and humidity ratio	Any
Scatter	Data explorer	Hourly	Any pair of variables	Any
Frequency plot	Data explorer	Custom	Any pair of variables	—

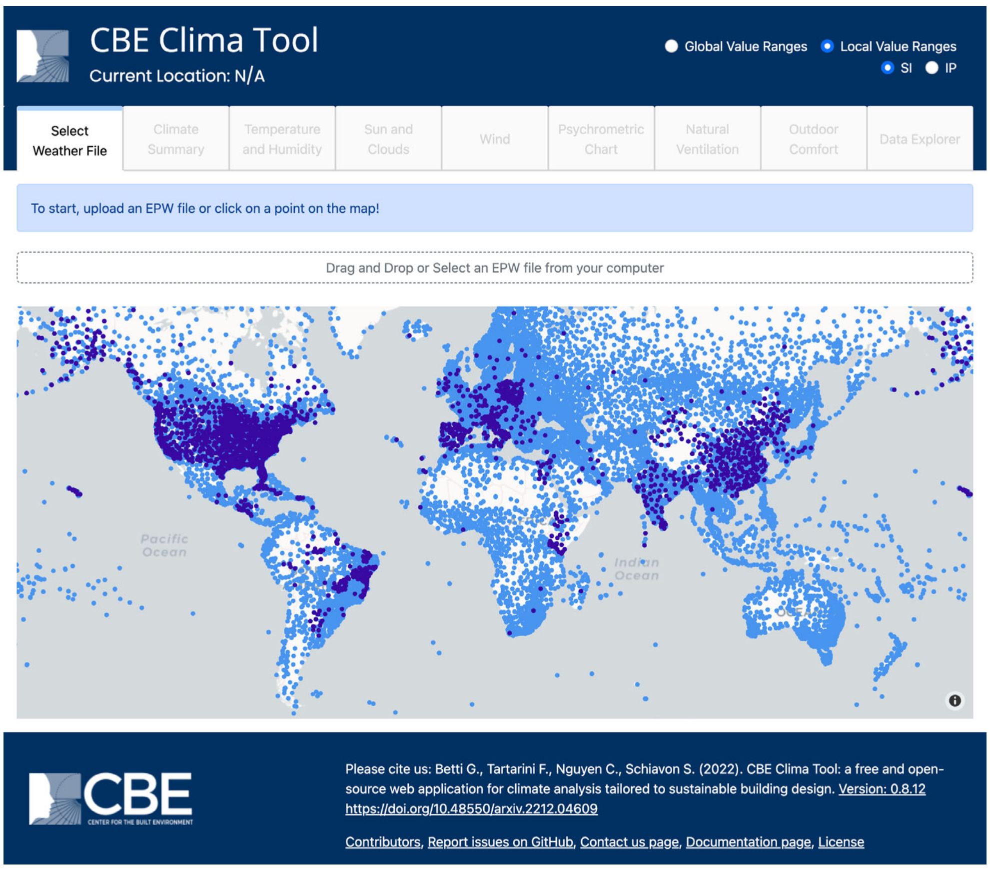

Fig. 3 CBE Clima Tool home page where the EPW file is selected or uploaded. On the top right is possible to select the units and if the data are visualized in a global (the same for every file) or local (unique to each file) range

axes as a function of the data contained in the EPW file. The user can also toggle between the International System (SI) and the Imperial System (IP) of units using the radio button located in the navigation bar.

The following paragraphs give an overview of the main functionalities of each page of the tool. Although there is a logical sequence in the organization of the tabs, they can be accessed in any order. We recommend consulting the official project documentation (https://cbe-berkeley.gitbook.io/clima/). This is a detailed and extensive description of the tool's functions. We think it is more effective to provide a detailed description in the online documentation given that there are no space constraints and it allows for updates every time new features are added.

Home Page Users can either choose to analyze the climate of the locations displayed on the map or upload a custom EPW file as shown in Figure 3. After loading an EPW file the user can then access the other tabs.

Climate Summary It provides textual information about the selected location, such as longitude, latitude, elevation, Koppen-Geiger climate zone, average/hottest/coldest yearly temperatures, annual cumulative horizontal solar radiation, and percentage of diffuse horizontal solar radiation. The user is presented with a further map showing the selected weather station's location and can click on the provided buttons to download the source data. The page displays the heating degree day (HDD) and cooling degree day (CDD) charts which are computed as a function of user-defined base temperatures. Charts depicting the distribution of the dry-bulb air temperature (T_{db}), relative humidity (RH), Global Horizontal Irradiance (GHI), and wind speed (U) are also displayed to the user on this page, as shown in Figure 4. Degree days are calculated as the integral of the difference between the outside air temperature and a base temperature over time. The user can define a base temperature (the outside temperature above which a building needs no heating or cooling) to be used to estimate degree days. The base temperature is not necessarily the desired building internal temperature but is the homeostatic temperature between the interior and exterior. The base temperature might depend, among other factors, upon the use of the building, the internal heat gains, and how resistant is the envelope to heat losses.

Temperature and Humidity It shows the yearly, daily,

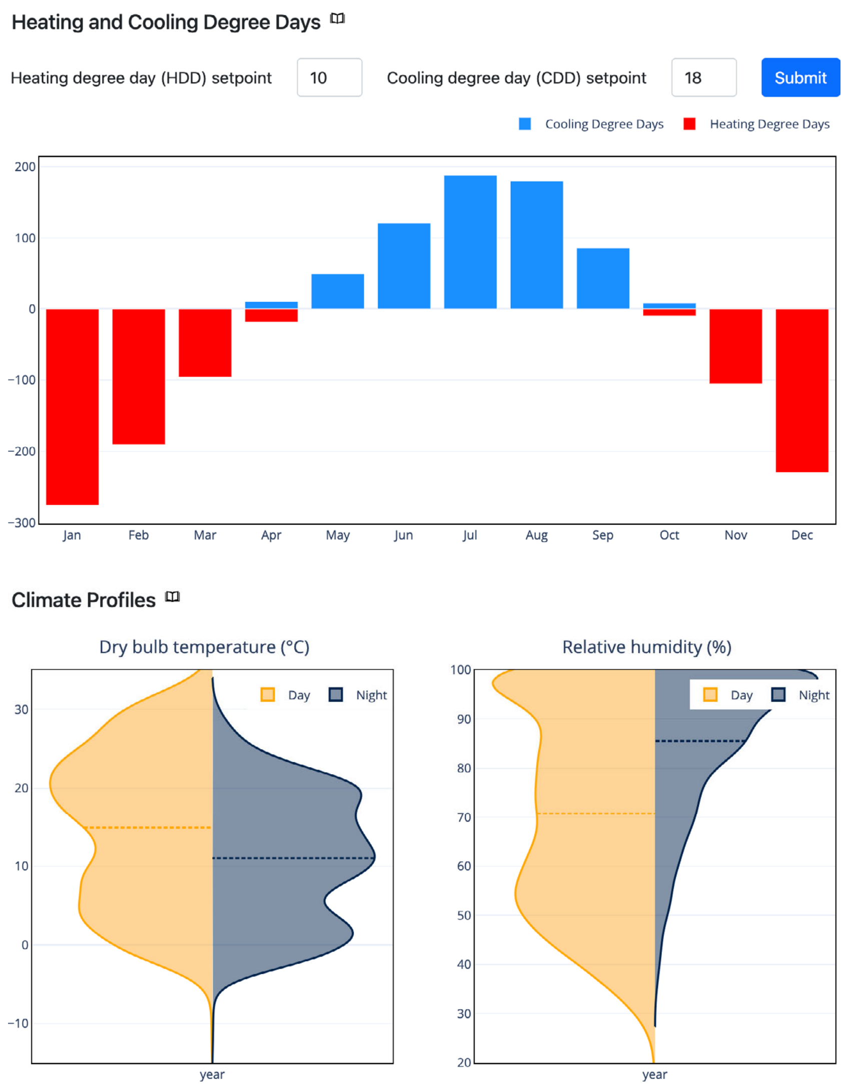

Fig. 4 CBE Clima Tool – Summary page – Heating and cooling degree and Climate profile charts for Bologna, Italy

and heat map charts for T_{db} and RH, as shown in Figure 5. The yearly T_{db} chart depicts the daily temperature range and the mean value together with the ASHRAE adaptive thermal comfort region for the 80% and 90% acceptability ranges. The prevailing mean outdoor temperature is calculated using an exponentially weighted running mean of the daily outdoor temperature for the last 7 days and assuming an α of 0.9 (CEN 2019). The daily chart shows how the hourly readings vary as a function of the hour of the day, and data are grouped by month. This chart also depicts the hourly

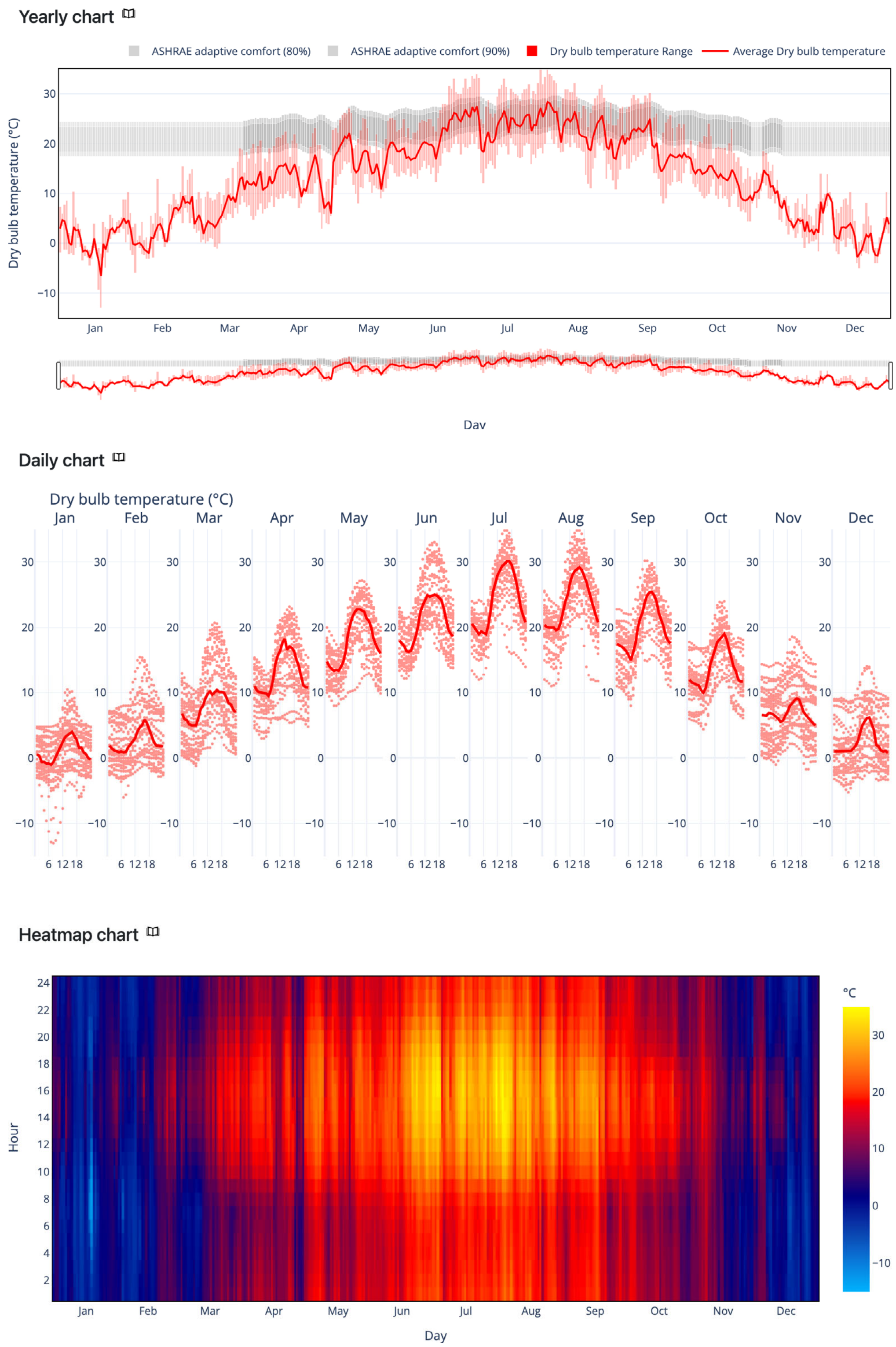

Fig. 5 CBE Clima Tool – Temperature and Humidity page – Yearly, daily, and heat map charts for Bologna, Italy

medians. The heat map shows each hourly data point in a grid where the vertical axis shows the time of the day, while the horizontal axis shows the day of the year. The table with descriptive statistics for each month and year is shown at the bottom of the page. This allows the user to get descriptive statistics for the selected location.

Sun and Clouds It visualizes the apparent sun path as seen by an observer at the weather station location either in Spherical coordinates (where the observer sits at the weather station location and the apparent altitude and azimuth are mapped to the polar and azimuthal angles respectively) or Cartesian coordinates (where the apparent altitude and azimuth are mapped to a Cartesian plot), as shown in Figure 6. The monthly average hourly global and diffuse solar radiation (kWh/m^2), and the monthly cloud coverage are also presented. The daily graphs showing the amount of energy gained from the sun have many uses, such as: Design the building with a passive approach to control solar gains and reduce energy consumption (allow solar gain during months of heating demand and block them during periods of cooling demand, see for reference the degree days).

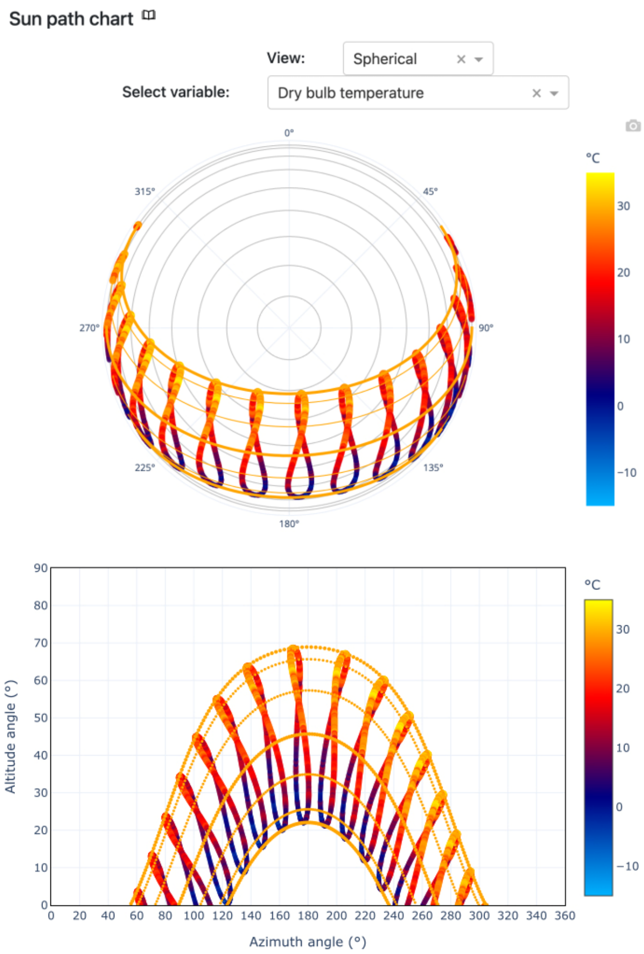

Fig. 6 CBE Clima Tool – Sun and Clouds page – Sun path in spherical and Cartesian coordinates for Bologna, Italy

Manage the direct solar gain through the glass, to evaluate solar shading devices (most useful in locations with high temperatures and a strong direct component). Manage the indirect solar gain transfer into the building with a time shift, exploiting the thermal mass, heating thick walls or concrete floors, or designing special rooms adjacent to the main spaces that rely on convection to transfer the heat, such as a sunroom or Trombe wall. Evaluating sustainable renewable energy solutions such as solar thermal or photovoltaic panels. The integral of the curves in the graphs is the total energy (in Wh/m^2) supplied by the sun. Users can also plot and visualize how any of the many variables related to solar radiation contained in the EPW file (e.g., GHI, extraterrestrial horizontal irradiation) vary throughout the year.

Wind It provides information about the wind velocity and direction for the selected location. It comprises several wind roses that either display the entire dataset (yearly data), values grouped by seasons or grouped by times of day. The wind rose is used to present the data since it provides a synthetic overview of both wind speed and direction frequency distribution at a given location. The user can also create a custom wind rose by filtering the data by month and hour of the day. This page also shows two heat maps, one for the wind speed and one for the wind direction, as shown in Figure 7. As with most graphs in Clima Tool, the wind rose is strongly interactive. Clicking on the legend will hide or highlight the selected category. As such, it is easy to go from a wind rose showing all the wind directions and frequency to one that highlights only the selected speed range. This can be particularly useful in identifying low-frequency, high-speed wind patterns.

Psychrometric Chart It shows data on a customizable psychrometric chart. Users can select to bin the data and color them by frequency or display the data and color them by any variable of choice. Optionally, filters can be applied to select data contained in a specific time range or filtered according to a secondary variable. As an example, see the figure in Section 3.2 for more. The psychrometric diagram allows users to understand the relationship between humidity and air temperature conditions. Through the use of the psychrometric diagram and appropriate calculations, it is possible to determine the amount of heat or cooling needed to achieve the desired temperature and humidity. The psychrometric diagram in the CBE Clima Tool shows T_{db} on the abscissae and the humidity ratio on the ordinates. Constant relative humidity lines are plotted as parametric curves inside the chart.

Natural Ventilation It allows the user to estimate when and for how long the climate is suitable for the use of natural ventilation, as shown in Figure 8. Natural ventilation is the process of introducing outdoor air into a building from the

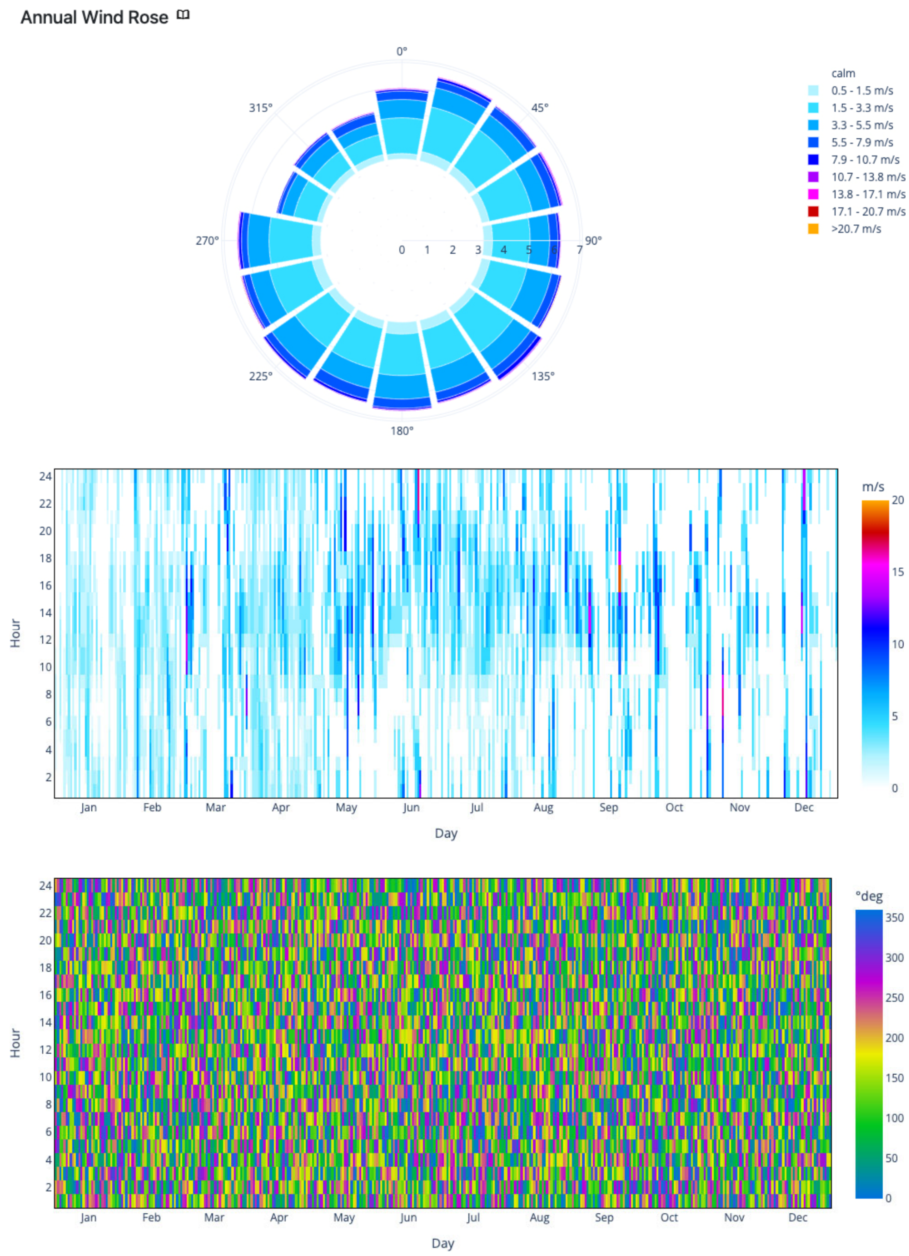

Fig. 7 CBE Clima Tool – Wind page – Annual wind rose, wind speed heat map, and wind direction heat map charts for Bologna, Italy

outside to improve air quality and provide fresh air and/or comfort cooling without actively conditioning the incoming airflow. In CBE Clima Tool the natural ventilation potential assessment is based primarily on the outdoor T_{db}. This can be achieved by selecting an outdoor T_{db} range in which natural ventilation can be used. The maximum temperature threshold can be estimated depending on the internal loads, how many people are in the room, and how many electrical appliances. The minimum temperature threshold is typically dictated by local discomfort near the fresh air inlet source. The user can also filter data either based on monthly or hourly ranges. The bar chart can be normalized, then displayed as a percentage of total monthly hours, or plotted with the total sum of the filtered hours. If radiant systems are used to cool the building, the user can also specify the surface temperature at which they operate. The CBE Clima Tool filters out automatically data when the surface of the radiant system is below the air temperature dew point.

Outdoor Comfort This page contains three charts showing the UTCI values calculated for the selected location, see Figure 9. The first heat map shows the UTCI equivalent temperature, while the second shows the data binned into the 10 UTCI stress categories (from extreme cold stress to extreme hot stress). The last chart shows the monthly percentage of time each of the 10 UTCI stress categories occurs in a specific month. The software calculated four

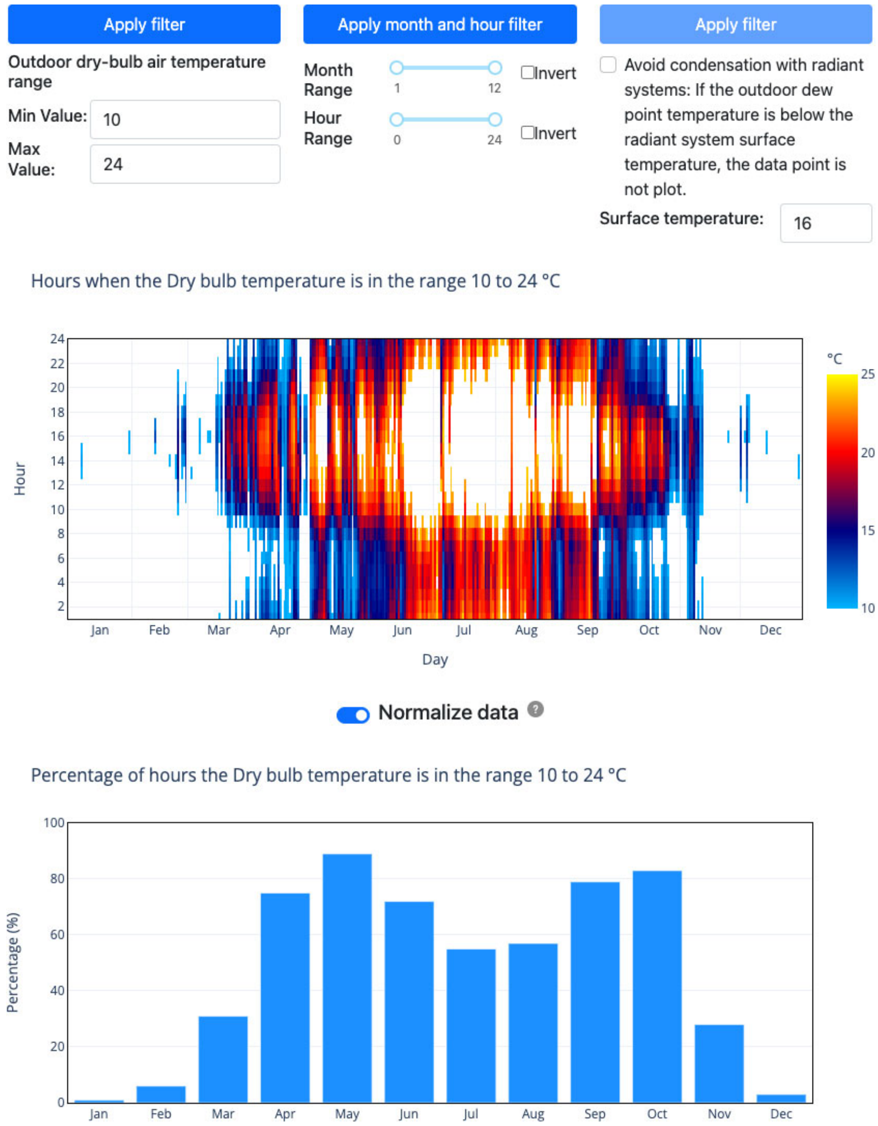

Fig. 8 CBE Clima Tool – Natural Ventilation page – Heat map and bar charts for Bologna, Italy

distinct UTCI scenarios showing the impact of exposure to the sun and the wind and combinations thereof. This might help the designer to understand what combination of exposure to or protection from the sun and wind would provide the most comfortable outdoor environment at different times of the year, this is particularly important in design because these two parameters are the most commonly modified to control the outdoor environment.

Data Explorer It allows the user to explore and plot all variables contained in the source dataset. It comprises three charts (yearly chart, daily chart, and heat map chart) that are also displayed on the "Temperature and Humidity" page. Moreover, it also allows users to visualize the correlation between three input variables. For example, Figure 10 is a screenshot of the aforementioned charts, and we are showing how data points are distributed as a function of T_{db}, RH, and GHI.

3 Illustrative examples

In this section, we will review a few possible use cases for Clima. The first two relate more to an educational setting, the third one is more representative of a possible use in a professional setting.

3.1 Sun path chart

Sun control and shading systems can significantly reduce peak cooling loads, while also improving the quality of natural lighting in the interior of the building and the thermal comfort close to the façade. This example shows how the user can visualize how T_{db} varies as a function of the position of the sun in the sky. This information can then be used to optimize the design of sun-shading devices. This chart can be generated from the "Sun and Clouds" tab.

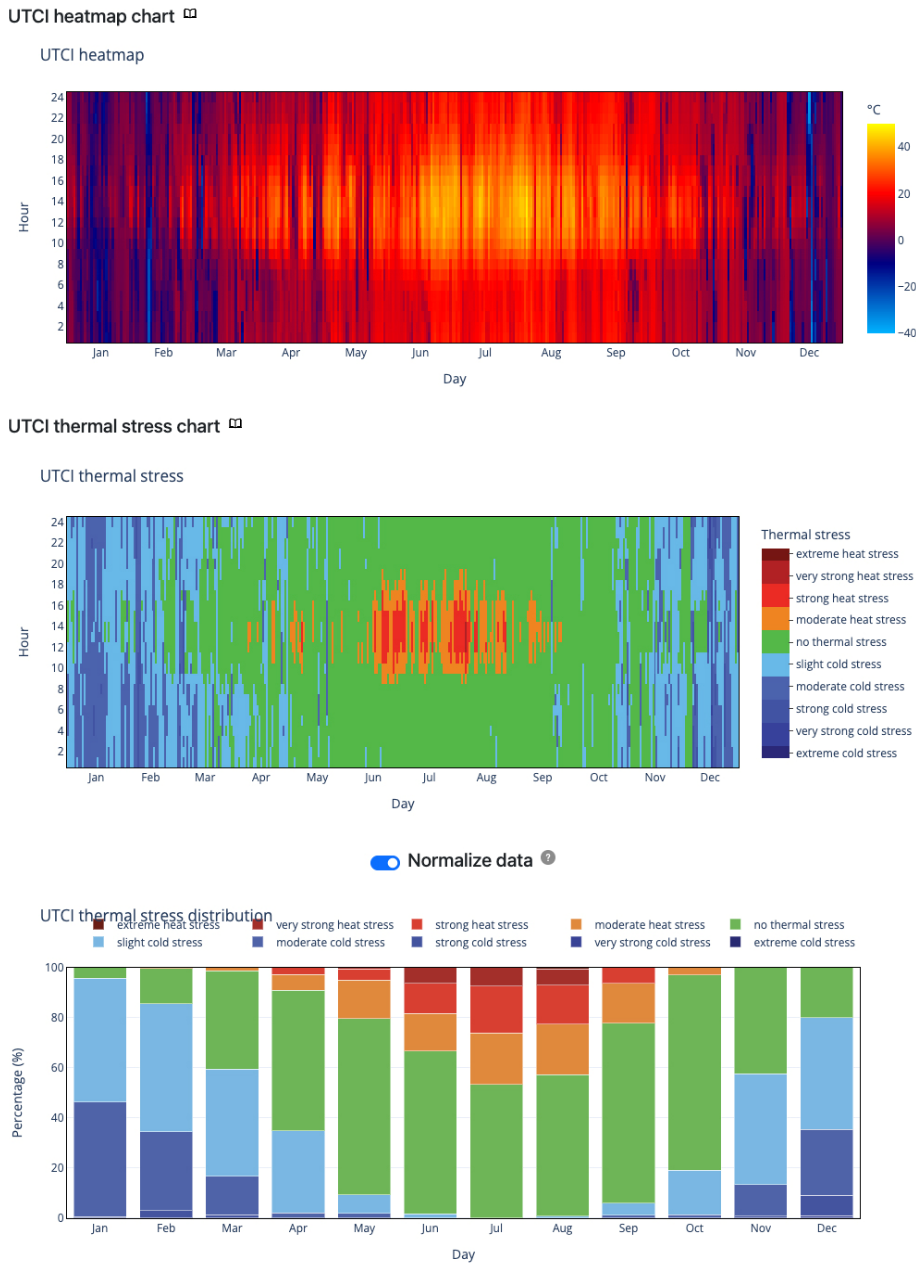

Fig. 9 CBE Clima Tool – Outdoor Comfort page – UTCI values heat map, UTCI stress category heat maps, and the monthly percentage of time of UTCI stress categories bar chart for Bologna, Italy

The user first has to select, using the drop-down, between a spherical or Cartesian coordinate system. In this example, we used the latter and decided to select T_{db} as a secondary variable to plot from the second dropdown. Figure 11 shows that this location is characterized by cold winters and hot summers. The user can then determine the optimal size and shape of the shading that can be used to minimize solar gains in summer while maximizing passive solar heating in winter.

3.2 Psychrometric chart

This example shows how the user can visualize the distribution of the climate data for a specific location on the psychrometric chart. We also decided to color the data points as a function of the UTCI calculated for the condition where the reference person is neither exposed to direct solar radiation nor the wind. This information can be used to design a semi-outdoor space or to determine when natural ventilation may be used. This chart can be generated from the "Psychrometric" tab. Through the drop-down located on the top left side of the page, the user can select the variable "UTCI: no Sun & no Wind". The CBE Clima Tool automatically generates the chart shown in Figure 12.

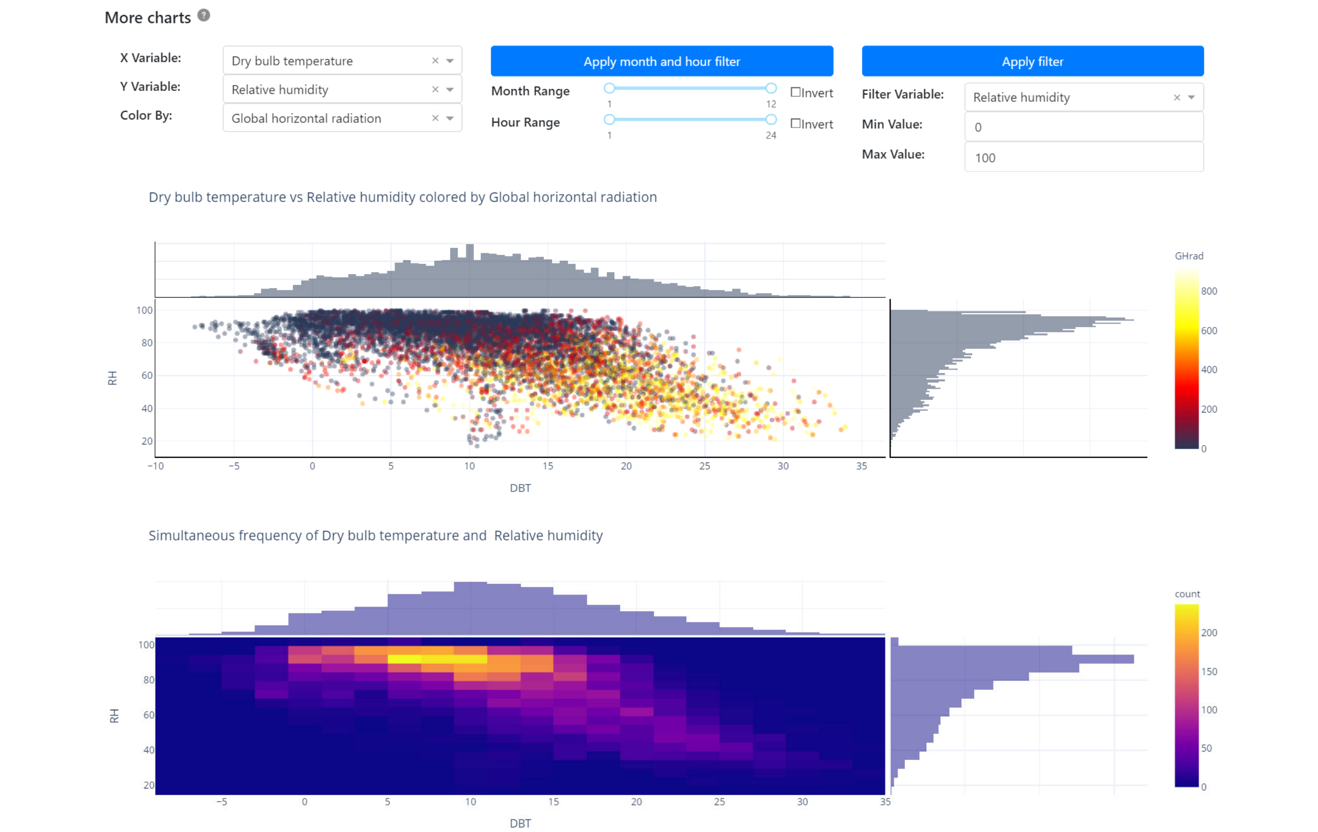

Fig. 10 Data explorer tab – The user can select two variables, one to be displayed on the *x*-axis and one to be displayed on the *y*-axis. This allows the user to visualize the correlation between the two variables. While the histograms depict the distribution of the data. This selection is then used to create a scatter plot and a heat map. In the scatter plot a third variable is used to color the markers and the range is displayed in the color bar. The heat map, on the other hand, depicts the correlation between the *x* and *y* variables and shows the number of points that are comprised in each bin

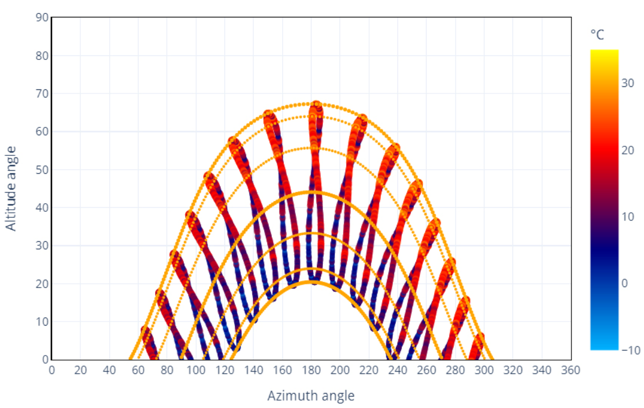

Fig. 11 Cartesian sun path chart. It can be used to determine the solar azimuth and altitude for every day lit hour of the year for a given location. In addition, we allow users to color the points using a third variable. This allows for example to determine how the T_{db} varies as a function of the solar azimuth and altitude throughout the year

3.3 Small office design

The following workflow imagines a designer tasked with developing a climate-adapted strategy for a small office in Oakland, California. A hypothetical yet representative workflow using Clima would start by locating the appropriate weather file on the interactive map or uploading the relevant EPW file for analysis. In this fictional example, the weather file USA CA Oakland.Intl.AP.724930 TMY3 is chosen as representative for the location. Subsequently, the user navigates to the "Climate Summary" tab to evaluate the heating and cooling degree days and understand the relative prevalence of heating or cooling in this climate. Based on the weather data and assuming a heating degree day (HDD) setpoint of 10 °C and a cooling degree day (CDD) setpoint of 18 °C, cooling predominantly influences the location. Intrigued to explore further, the user decides to experiment with the sensitivity of this metric by adjusting the HDD setpoint to 15 °C. This adjustment results in a reversal of the graph, indicating the dominance of heating requirements. Recognizing that temperatures frequently fall between 10 °C and 15 °C in the area, the user acknowledges the significance of reducing the building's homeostatic temperature to around 10 °C. This can be accomplished by improving the insulation of the building envelope and maximizing solar gains during winter (Figure 13).

The user can then proceed directly to the "Natural Ventilation" tab to estimate the periods when natural ventilation is feasible. Since the hypothetical example involves designing an office building, the user chooses to view the results only between 7 a.m. and 7 p.m. The outcomes are promising. Natural ventilation (for outdoor temperatures between 10 °C and 24 °C) appears viable for most of the year. The monthly graph demonstrates that the range typically fluctuates from a low of 72% of the hours in January to 100% in July. Considering the relatively high discomfort risk near windows when introducing air at 10 °C, the user decides to investigate the impact of limiting ventilation to times when the outdoor temperature ranges from 14 °C to 24 °C. This adjustment significantly reduces the suitable hours for natural ventilation but still maintains more than 60% of occupied time from April to November. Equipped with this information, the user proceeds to check the compatibility of the climate with radiant systems and is pleasantly surprised to find a minimal risk of condensation (Figure 14).

Next, the user examines the "Wind" tab to understand the speed and seasonal direction of the wind rose for the location, aiding in the planning of opening orientations (Figure 15).

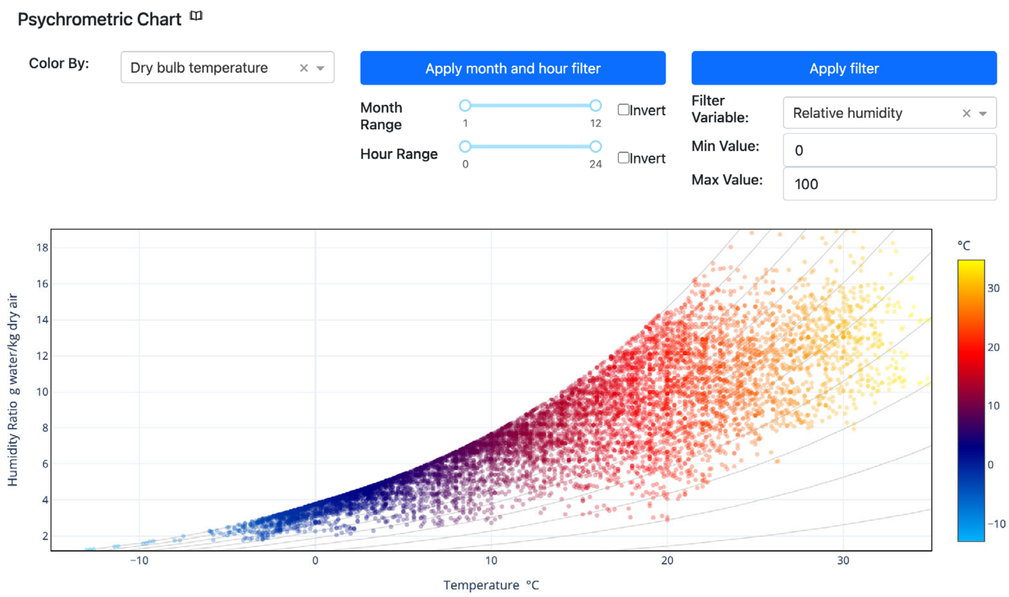

Fig. 12 Psychrometric chart. It can be used to visualize the relationship between dry air and moisture. In addition, we allow users to color the points using a third variable, in this example, we are showing how the UTCI – No Sun and Wind – varies as a function of the dry-bulb air temperature and the humidity ratio.

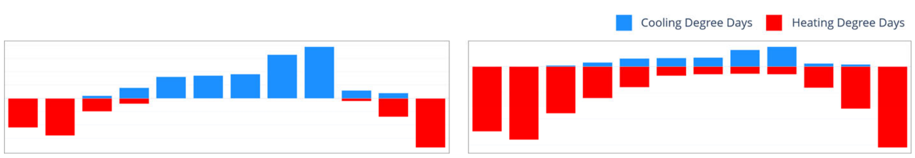

Fig. 13 Cooling degree days for Oakland, CA, USA. The graph on the left heating degree day (HDD) setpoint of 10 °C and a cooling degree day (CDD) setpoint of 18 °C, and the graph on the right has a heating degree day (HDD) setpoint of 15 °C and a cooling degree day (CDD) setpoint of 18 °C

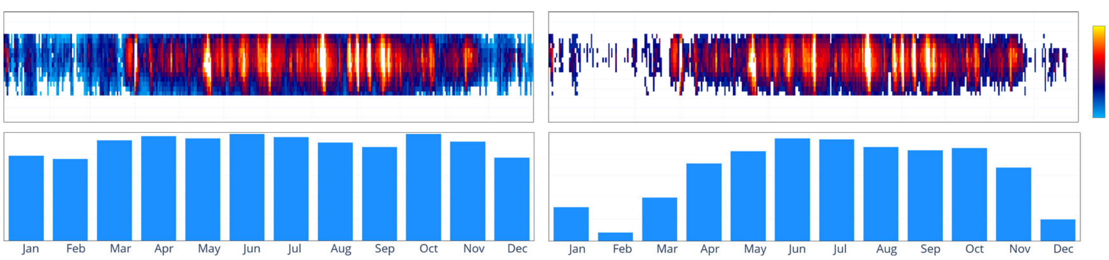

Fig. 14 Natural ventilation potential for Oakland, CA, USA, between the hours of 7 a.m. and 7 p.m. Ventilation is assumed possible while outdoor temperatures are between 10 °C and 24 °C (left) and 14°C and 24 °C (right)

Finally, in our hypothetical scenario, the user intends to design a small outdoor lunch area for future office occupants. To accomplish this, they access the "Outdoor Comfort" tab and select the time range between 11:00 and 15:00. By reviewing the four available scenarios (Sun & Wind, Sun & no Wind, no Sun & Wind, no Sun & no Wind), the user discovers that the most comfortable conditions are achieved in an area shielded from both sun and wind. Leveraging their familiarity with the wind patterns in the area, the hypothetical user then consults the "Sun and Clouds" tab to gain a better understanding of the sun path for the location and the frequency of sunny, partially cloudy, and overcast skies. This information assists in the effective planning of the outdoor space (Figure 16).

4 Discussion

The CBE Clima Tool is intended to be used by architects, engineers, researchers, educators, and students. It can even be used by non-technical users with no analytical or programming skills. Compared with similar competing solutions for climate analysis, CBE Clima Tool offers a complete and flexible feature set that makes it useful for a varied group of users. The CBE Clima Tool can be used by students exposed to this topic for the first time and by designers with many years of experience. The easy output of high-quality vector graphics licensed under Creative Commons Attribution 4.0 International (CC BY 4.0) facilitates its integration into presentation documents.

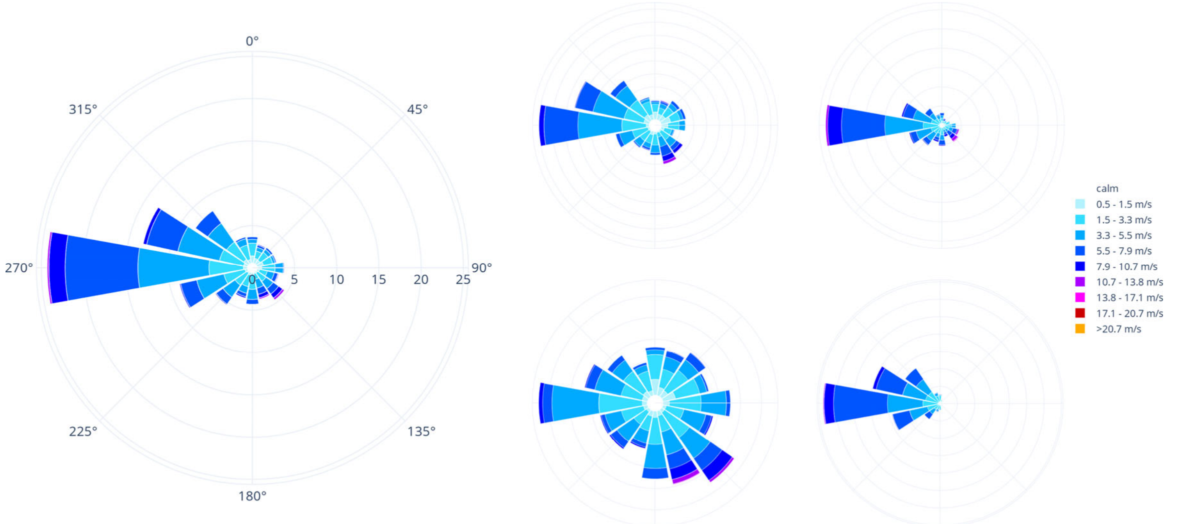

Fig. 15 Wind roses for Oakland, CA, USA. On the left, in large, the annual wind rose, on the right, small, from top right moving clockwise: Fall, Spring, Summer and Winter

Fig. 16 Universal Thermal Climate Index (UTCI) for Oakland, CA, USA, under various conditions (from top right, clockwise) exposed to sun and wind, exposed to sun and shielded from wind, shielded from sun and wind, shielded from sun and exposed to wind

While alternative solutions require either subscriptions or one or more installs, CBE Clima Tool can be used via any device equipped with a browser. Some of the most unique features of the tool, in addition to the completeness, interactivity and customizability of the data visualizations produced, pertain to the availability of complex metrics such as Natural Ventilation potential and the Universal Thermal Climate Index (UTCI) under four distinct scenarios. We have also open-sourced the code so users around the world can contribute to our project or use our functions in their applications. This also allows users to run the code locally, if access to the online tool is censored by the country in which they are living. The source code is licensed under the MIT license. This permissive license enables users to share, copy, and redistribute the material in any medium or format and to adapt, remix, transform, and build upon the material for any purpose, even commercially. We decided to use this permissive licence to promote the widest possible adoption. We released the first version of this tool in August 2021. Since its release, the tool has consistently gathered over 2,000 unique monthly users from more than 70 nations worldwide.

5 Conclusions

The CBE Clima Tool is a free and open-source web application for analysing and visualising climate data specifically designed to support the needs of architects, engineers, students, and educators interested in the design of climate-adapted buildings. It aims to reduce barriers to the implementation of sustainable design. CBE Clima Tool processes and visualizes EPW climate files. It allows users to perform complex psychrometric data calculations and interactively visualize the results without the need to write any code or install specialized software. It is written in Python. All generated charts can be downloaded in SVG format; this allows them to be modified and adapted according to the users' needs. It is composed of the following main pages: home, climate summary, temperature and humidity, sun and clouds, psychrometric chart, natural ventilation, outdoor comfort, and data explorer. The pages contain interactive graphs and tables. The tool allows an easy and accessible interpretation of climate data.

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Declaration of competing interest

We confirm that there are no known conflicts of interest associated with this publication and that there has not been financial support for this work that could have influenced its outcome.

Author contributions statement

Giovanni Betti: conceptualization, methodology, software, investigation, writing—reviewing and editing, visualizations, supervision, project administration, funding acquisition. Federico Tartarini: methodology, software, validation, formal analysis, data curation, writing—original draft, writing—reviewing and editing, visualization, project administration. Christine Nguyen: methodology, software, formal analysis, data curation, visualization. Stefano Schiavon: conceptualization, writing—reviewing and editing, supervision, funding acquisition.

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